

Mikhail Razakov

Compiler Tools Developer

misha13022008@gmail.com · t.me/Micodiy · github.com/Misha1302

Moscow / remote
from September 2026
up to 20 h/week

In UniversalToolchain/Wist2, I built a typed API, structured diagnostics, a versioned trace, parity/negative tests, and package smoke checks. For the PS-form Analyzer, I built a generator and test harness with an exact oracle, metamorphic tests, and WA/TL hunting modes.

8 additional defects

found by a second verification pass and covered by regressions

1,325 tests

parity, negative, lifecycle, and architecture checks

Exact oracle + metamorphic tests

case generation, WA/TL hunting, and counterexample retention

EXPERIENCE

2024 — present

UniversalToolchain / Wist2 — Creator and Primary Developer

Compiler SDK · diagnostics · verification infrastructure

- Built a typed facade, structured diagnostics, and a versioned redacted trace for the public Wist API.
- Added interpreter/CIL parity tests, negative contract tests, lifecycle checks, and architecture guard tests.
- Added package-surface and clean-consumer smoke tests; the public baseline covers 75 .NET projects and 1,325 tests.

2026

PS-form Analyzer — Test Harness and Solver Verification

Python generators · exact oracle · metamorphic testing

- Built a Python harness with fixed and generated cases, strict stdout validation, and per-case timeouts.
- Added an exact oracle for small domains plus randomized and metamorphic tests.

SKILLS

RECOGNITION

Programming olympiads

Prize-winner of the HSE "Vysshaya Proba" Olympiad in competitive and industrial programming; prize-winner of the regional Russian Informatics Olympiad.

Compiler projects

Two highest-tier diplomas in the MEPhI Junior competition and a highest-tier diploma in the Baltic Science and Engineering Competition.

EDUCATION

HSE University student

SELECTED PROJECTS

.NET · SDK · compiler/runtime tooling

UniversalToolchain / Wist2

A compiler/runtime framework with a public API, two execution backends, and dedicated testing infrastructure.

- structured errors, versioned traces, and explanation APIs.
- interpreter/CIL parity, structural verifiers, negative contracts, and state-isolation tests.

Python · generative testing

PS-form Verification Harness

A generator and runner for a conservative memory-dependence analyzer.

- Exact oracle for small domains and fixed regressions.
- Randomized and metamorphic generation.

C++23 · differential testing

AdvancedAlgorithms Verification Infrastructure

Validation for 12 reusable modules through reference algorithms, invariants, invalid-input cases, and large inputs.

- Differential oracles for max flow, SCC, Dijkstra, HLD, and other modules.
- ASan/UBSan, separate correctness/performance CTest labels, and API-contract checks.